

Intelligent Detection of Abnormal Traffic in Mobile Switching Centres

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Abstract

This project classifies traffic in Mobile Switching Centres into three categories: Normal, Abnormal, and Drone anomaly. The classification is done using NPS parameters as the primary layer, because parameters like average packet loss, delay, and throughput directly reflect the operating state of the MSC under different load conditions.

A dataset of 260 samples was manually collected using Cisco Packet Tracer across 13 scenario groups. The `ping -n 50` command was run on each active device for each simulation run, and the average RTT and lost packet count were recorded. These values were used to compute `avg_delay` and `packet_loss`, which are the two most discriminating features in the dataset according to feature importance analysis.

NPS parameters alone cannot detect drone anomalies, because a drone produces moderate packet loss and delay values that overlap with Normal conditions. The project extends the NPS base using Electromagnetic Wave Theory and Transmission Lines parameters, specifically Doppler shift and fading rate, which change distinctly when a fast-moving aerial node is present. Three classifiers were trained: Random Forest, SVM, and KNN. Random Forest and SVM achieved 100% accuracy on the test set.

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Introduction

1. The Mobile Switching Centre is the central node in a cellular network, and all calls, data sessions, and signalling messages pass through it. When too many devices send traffic simultaneously, the input queue at the MSC fills up and packets are dropped, which is called a signalling storm. Detecting this condition early is important because it can make the entire network unreachable for connected devices.
2. This project was originally designed to classify MSC traffic into Normal or Abnormal using basic NPS parameters such as average packet loss, delay, and throughput. These parameters are measurable directly from a Cisco Packet Tracer simulation, so the dataset was built by running ping sessions and recording the output for each scenario.
3. NPS parameters cannot detect drone movement, because a drone produces moderate packet loss and delay values that look similar to lightly loaded Normal conditions. To address this, the project was extended to drone anomaly detection using EMTL parameters. If ISAC is implemented on existing base stations, the same signals used for communication can also be used for sensing, so no additional hardware is required for drone detection.
4. The NPS parameters form the base for primary classification of Normal versus Abnormal traffic. The EMTL and Wireless parameters extend the scope to drone anomaly detection, which is a separate problem that the NPS layer alone cannot solve. Together the two layers cover all three traffic classes using a single 260-sample dataset collected from Cisco Packet Tracer.
5. Three machine learning classifiers were trained on the combined 21-feature dataset: Random Forest, SVM, and KNN. Random Forest and SVM both achieved 100% accuracy on the held-out test set. No classifier produced a False Negative, meaning

no actual Abnormal or Drone sample was labelled as Normal in any of the three models.

Methodology

The methodology covers four areas: the NPS theoretical model, the Cisco Packet Tracer setup, the data collection procedure, and the machine learning pipeline.

1. NPS Theory

Traffic intensity is the ratio of the packet arrival rate to the router's service rate, expressed as $\rho = \lambda / \mu$. When ρ is below 0.7, the queue at Router0 is stable and delay stays low. As ρ approaches 1.0, the queuing delay grows according to $W_q = \rho / (\mu (1 - \rho))$, which means a small increase in traffic can cause a large jump in delay. This non-linear growth is the theoretical basis for the delay explosion seen in Abnormal scenarios.

Total one-way packet delay is the sum of queuing delay, transmission delay (1500 times 8 divided by throughput in bps), and propagation delay (approximately 5 ms per LAN link). In Normal scenarios this stays between 0.02 and 0.55 seconds. In Abnormal scenarios at ρ of 0.92 to 0.99, it rises to between 2.5 and 4.2 seconds.

Packet loss occurs when the input buffer at Router0 overflows and newly arriving packets are discarded. It is calculated as the number of lost packets divided by 50, taken directly from the ping output. This is the most discriminating NPS feature in the dataset, with a feature importance score of 0.1713.

Throughput and traffic rate behave differently under congestion. The router keeps receiving packets at a high rate, but most are dropped before being forwarded, so the effective throughput at the server collapses. This is why the show interfaces command on Router0 shows a Low or Very Low output rate in Abnormal scenarios even though devices are still sending.

2. Cisco Packet Tracer Topology

The following topology was used for all 260 simulation runs. Only the number and type of active devices changed between runs.

```
PC0  --+
PC1  --+
PC2  --+-- Switch0 -- Router0 (MSC) -- Switch1 -- Server0
PC3  --+                                     +--
Server1
PC4  --+
AP0  --- Laptop0, Smartphone0
AP1  --- Laptop1, Smartphone1
```

Router0 acts as the simulated MSC. All traffic from all devices passes through it, so increasing the number of active devices directly increases the load on Router0. IP addresses were assigned manually in Realtime mode using Desktop > IP Configuration on each device and the CLI on Router0.

3. Data Collection Procedure

For each of the 260 simulation runs, the same procedure was followed.

- Set the number of active PCs for the run. Open Command Prompt on each active device.
- For Group A2 runs: enter Router CLI, type configure terminal, then interface fa0/0, then bandwidth 128 or bandwidth 256.
- Run the ping command on each active device: ping 192.168.2.10 -n 50
- Record Average RTT and Lost count from the ping output. Compute avg_delay as Average RTT divided by 2000 (to convert milliseconds to seconds). Compute packet_loss as Lost divided by 50.

- Run show interfaces fa0/0 on Router0. Read the Output rate field and assign a throughput label: High, Medium, Low, or Very Low.
- Assign traffic_intensity based on device count and traffic type. Assign the class label using the thresholds in Table 2.
- If bandwidth was throttled, type no bandwidth on the interface to reset it before the next run.

CPT has two operating modes. Realtime mode is the default and is where all setup and data collection was done. Simulation mode pauses time and shows each packet as an envelope on the canvas. Simulation mode was used for visual verification: during Abnormal runs, packets could be seen disappearing at Router0, confirming that tail-drop was occurring at the MSC bottleneck rather than elsewhere. The Event List panel showed each packet's status at every hop.

4. Scenario Groups

Table 1: Simulation Scenario Groups

Group	Scenario	Runs	Label	Condition
N1	Single client, idle	15	Normal	PC0 pinging Server0 only. Baseline clean condition.
N2	Two clients, light load	15	Normal	PC0 and PC1 pinging simultaneously. Light shared load on Router0.
N3	Wireless, good signal	15	Normal	Laptop close to AP (green signal). Stable wireless baseline.
N4	Mixed wired and wireless	15	Normal	2 PCs plus 1 wireless device. Balanced mixed traffic.
N5	Three wired PCs	10	Normal	PC0, PC1, PC2 pinging together. Approaching load boundary.
A1	Signalling storm	20	Abnormal	All 5 PCs plus wireless devices flooding Server0 simultaneously.

Group	Scenario	Runs	Label	Condition
A2	Throttled router	15	Abnormal	Router throttled to 128 or 256 kbps via CLI bandwidth command.
A3	Large-packet flood	15	Abnormal	Traffic Generator: max packet size, 0.1 second interval, 3 to 4 PCs.
A4	Dual-server saturation	10	Abnormal	PC0 to PC2 targeting Server0, PC3 and PC4 targeting Server1.
A5	All-device overload	10	Abnormal	All 9 devices (5 wired, 4 wireless) transmitting at once.

Table 2: Label Assignment Thresholds

Metric	Normal	Abnormal	Drone
avg_delay	Less than 0.60 s	More than 0.70 s	0.28 to 1.20 s
packet_loss	0.00 to 0.02	0.05 and above	0.03 to 0.12
throughput	Medium or High	Low or Very Low	Medium
active_pcs	1 to 5	6 to 15	0 to 3

5. Machine Learning Pipeline

- Encoding: ordinal variables like traffic_intensity and throughput are mapped to integers that preserve their natural order. Nominal variables like traffic_type are label-encoded.
- Scaling: StandardScaler is applied to all 21 features to bring them to mean 0 and standard deviation 1. This is necessary because SVM and KNN are distance-based and would otherwise give too much weight to features with large numerical values.
- Split: 208 training samples and 52 test samples, using stratified sampling to maintain the same class distribution in both sets.
- Cross-validation: 5-fold stratified cross-validation on the training set, using f1_macro as the scoring metric.

- Models: Random Forest with 200 trees and class_weight balanced, SVM with RBF kernel and one-vs-rest strategy, KNN with K=5 selected by GridSearchCV from the values 3, 5, 7, 9, and 11.

Experiments

Experiments were run across all 13 scenario groups using the same topology. The variable between experiments was the number and type of active devices. Router0 bandwidth was additionally throttled for Group A2, and wireless device position was varied for the Drone groups.

Normal Class - Groups N1 to N5 (70 samples)

- N1: PC0 only pinging Server0. Baseline condition. avg_delay stayed between 0.02 and 0.15 seconds, packet_loss was 0.00 to 0.01, and throughput was High throughout.
- N2: PC0 and PC1 simultaneously. Light shared load on Router0. avg_delay increased slightly to 0.10 to 0.35 seconds but no packet loss occurred.
- N3: Laptop close to AP with green signal indicator. avg_delay was 0.05 to 0.20 seconds, no loss, SNR above 25 dB throughout.
- N4: Two PCs and one wireless device active together. avg_delay stayed below 0.55 seconds with minimal loss.
- N5: Three PCs pinging simultaneously. avg_delay reached 0.30 to 0.55 seconds, with occasional 1 to 2 percent loss at the upper end.

Abnormal Class - Groups A1 to A5 (130 samples)

- A1: All 9 devices flooding Server0. This is the most severe scenario. avg_delay was 2.5 to 4.2 seconds, packet_loss was 0.28 to 0.50, and throughput was Very Low.
- A2: Router bandwidth throttled to 128 kbps. Even 2 active PCs caused congestion at this limit. avg_delay was 1.5 to 3.5 seconds and packet_loss was 0.15 to 0.40.

- A3: Traffic Generator set to maximum packet size and 0.1 second interval. Large packets increase per-packet processing time at Router0, so congestion occurred at lower device counts than A1.
- A4: Split flood with two separate targets. Both paths through Router0 became congested, producing avg_delay of 1.8 to 3.2 seconds on both sides.
- A5: All devices transmitting simultaneously. Produced the most extreme metric values in the Abnormal class.

Drone Class - Groups D1 to D3 (60 samples)

- D1 (20 runs): Single laptop at the far corner of the canvas, 0 to 2 background PCs. avg_delay was 0.28 to 0.65 seconds and packet_loss was 0.03 to 0.08. These values overlap with Normal, which is why NPS parameters alone cannot identify this class.
- D2 (20 runs): Laptop at the midpoint between AP0 and AP1. The device oscillated between access points, producing 6 to 10 handovers per session and packet loss from the brief service gaps during re-association.
- D3 (20 runs): Two wireless devices at AP coverage edges with no background wired traffic. Handover count was 7 to 12. The channel variation was the highest across all Drone runs because both devices were simultaneously at their coverage boundaries.

Key Observations

- The M/M/1 model prediction was confirmed experimentally. When active_pcs increased from 5 to 6, avg_delay jumped by a factor of 3 to 5, matching the non-linear growth of $W_q = \rho \text{ divided by } (\mu \text{ times } (1 \text{ minus } \rho))$.

- Drone packet_loss of 3 to 12 percent overlaps with the lower end of Abnormal values, confirming that these two classes cannot be separated using packet_loss alone.
- Simulation mode visual observation confirmed that dropped packets appeared at Router0 during Abnormal runs, not at the Switch or Server, which validates that the M/M/1 model is the correct framework for this scenario.

Results

Model Performance

Table 3 shows the performance of all three classifiers on the 52-sample test set. All metrics use macro-averaging, so each class contributes equally to the final score regardless of how many samples it has.

Table 3: Model Performance on Test Set (52 samples, macro-averaged)

Model	Accuracy	Precision	Recall	F1-Score	CV F1
Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000
SVM (RBF Kernel)	1.0000	1.0000	1.0000	1.0000	1.0000
KNN (K=5)	0.9808	0.9877	0.9762	0.9814	0.9955

Random Forest and SVM both achieved 100% accuracy on the test set. KNN achieved 98.08%, with one Normal sample misclassified as Abnormal. No classifier produced a False Negative, meaning no actual Abnormal or Drone sample was labelled Normal. In a monitoring context this matters more than accuracy, because missing a real overload event is worse than generating a false alarm.

Feature Importance

Table 4 shows the top 10 features by importance score from the Random Forest model, measured as mean decrease in Gini impurity across 200 trees.

Table 4: Top 10 Feature Importances (Random Forest)

Feature	Importance	Layer	What it indicates
packet_loss	0.1713	NPS	Primary overload indicator. Rises from near zero to 50% under a signalling storm.
active_pcs	0.1404	NPS	Root cause of overload. Sets the arrival rate in the M/M/1 queuing model.
fading_rate	0.1108	EMTL (extension)	Primary drone indicator. Fast-moving nodes produce fading rates of 0.55 to 0.80.
velocity_mps	0.1046	EMTL (extension)	Drone velocity of 5 to 30 m/s is not reachable by walking users or stationary PCs.
channel_variation	0.1012	Wireless	Captures rapid channel amplitude change caused by drone mobility.
avg_delay	0.0742	NPS	Shows the M/M/1 queue effect. Normal: 0.02 s. Saturated MSC: 4.20 s.
handovers	0.0569	Wireless	Drones cross AP boundaries repeatedly, producing 6 to 12 handovers per session.
fading_depth_db	0.0547	EMTL (extension)	Multipath fading depth increases with node velocity and interference level.
interference_dbm	0.0409	EMTL (extension)	Co-channel interference rises with active device count, marking Abnormal scenarios.
freq_variation_hz	0.0359	EMTL (extension)	Carrier drift caused by Doppler effect. Secondary confirmation of drone motion.

The results split into two clear groups. The NPS group, which includes packet_loss, active_pcs, and avg_delay, accounts for 38.6 percent of total importance and is sufficient to classify Normal versus Abnormal traffic. The EMTL and Wireless group, which includes fading_rate, velocity_mps, channel_variation, and handovers, accounts for 38.4 percent and is responsible for separating the Drone class from the other two.

This confirms the core idea of the project. NPS parameters handle the primary classification problem. EMTL parameters extend the scope to drone detection, which NPS cannot do on its own. Neither group alone would be sufficient for all three classes.

Dashboard Observations

- Average Delay panel: Normal samples cluster between 0.02 and 0.55 seconds. Abnormal samples spread from 0.70 to 4.20 seconds. The jump at the 0.70 second threshold is sharp, which matches the M/M/1 non-linear prediction. Drone samples sit between the two, which is why delay alone cannot separate them from low-severity Abnormal cases.
- Packet Loss panel: Normal median is near zero. Abnormal median is 0.22. Drone median is 0.07. The overlap between Drone and low-severity Abnormal at the lower end is visible in the boxplot, and this is the quantitative reason the EMTL extension was added.
- PCA panel: All three classes form separate clusters in the 2D reduced space, confirming that the 21 features together are sufficient to discriminate all three classes even though individual NPS features have overlapping ranges between Drone and the other classes.

Conclusion and Future Work

This project classifies Mobile Switching Centre traffic into Normal, Abnormal, and Drone categories using NPS parameters as the primary layer and EMTL parameters as an extension. The NPS layer handles the primary classification problem, which is detecting MSC overload from signalling storms. The EMTL layer extends the scope to drone anomaly detection, which NPS parameters cannot support because drone scenarios produce metric values that overlap with Normal conditions.

The dataset of 260 samples was collected manually from Cisco Packet Tracer using the ping -n 50 command and the show interfaces CLI output. The M/M/1 queuing model correctly predicted the non-linear delay increase observed in Abnormal scenarios, confirming that the theoretical framework is appropriate for this simulation setup. Feature importance analysis showed that packet_loss and active_pcs are the two most important features, both of which are direct NPS observables.

Random Forest and SVM both achieved 100% accuracy on the test set. KNN achieved 98.08%. No classifier produced a False Negative across any of the three models. The feature importance results confirm that NPS features dominate MSC overload detection while EMTL and Wireless features dominate drone detection, which validates the layered design.

If ISAC is implemented on existing base stations, the same signals used for communication can be repurposed for sensing, so drone detection does not require separate hardware. The Doppler shift and fading rate features identified in this project are exactly the signals that an ISAC system would use for aerial node detection. Future work can extend this to real RF measurements using Software Defined Radio equipment, add a jamming or DDoS class as a fourth category, and deploy the trained classifier as a real-time monitoring agent.

References

1. A. Erlang, "The theory of probabilities and telephone conversations," *Nyt Tidsskrift for Matematik B*, vol. 20, pp. 33-39, 1909.
2. L. Kleinrock, *Queueing Systems, Volume 1: Theory*. Wiley-Interscience, 1975.
3. T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed. Prentice Hall, 2002.
4. S. Haykin, *Communication Systems*, 4th ed. Wiley, 2001.
5. D. Tse and P. Viswanath, *Fundamentals of Wireless Communication*. Cambridge University Press, 2005.
6. Cisco Systems, "Cisco Packet Tracer User Guide," Version 8.x, 2023.
7. 3GPP TS 23.501, "System architecture for the 5G System (5GS)," Release 16, 2020.
8. L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5-32, 2001.
9. C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, pp. 273-297, 1995.
10. T. Cover and P. Hart, "Nearest neighbor pattern classification," *IEEE Transactions on Information Theory*, vol. 13, no. 1, pp. 21-27, 1967.
11. F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825-2830, 2011.
12. A. Liu et al., "A Survey on Fundamental Limits of Integrated Sensing and Communication," *IEEE Communications Surveys and Tutorials*, vol. 24, no. 2, pp. 994-1036, 2022.